# Advances in Data Mining for Software Engineering

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**Abstract:** Analysis on application of data mining to software engineering projects indicates that the information rules mining technique has been used in many fields and succeeds in analyzing and organizing a large amount of data. Meanwhile, the information rules mining technique extracts useful information and displays it to the people. In this context, we analyze the latest advances in data mining that are applied to software engineering in this paper, explore the reason behind the widespread application of the data mining technique, and discuss the challenges that occur during the development. At the same time, we look into the future of data mining to gain remarkable insights into this technique.
**Key words:** software engineering; data mining; research advances

## 1. Introduction

With increase in the amount of data involved in the software engineering projects, it is more challenging to analyze and learn the data accumulated during software development using the traditional methods. Meanwhile, in the context of continuous advances in computer science and data mining for software engineering projects, the studies about data mining will continue to focus on the development of intelligent and automatic techniques for the purpose of tapping the potential of data mining and adapting to larger-scale data systems. By enhancing the technological analysis tools, the data mining technique can be used more effectively for software engineering.

Data mining refers to the process of extracting useful knowledge from the large amount of data. Software data mining is a concept that has close relation with software engineering and computer application techniques. It adopts a novel calculation method to find the desired data from the large amount of data. That is, the knowledge which is useful to software developers is found out using existing techniques to perform data extraction, analysis and representation. Essentially, this is a novel way of data analysis. The modern software engineering projects are very large in scale, and the dramatic increase in the code lines of software has resulted in rapid increase in the data volume. The traditional data calculation patterns are thus unable to meet the modern requirements on data analysis. In this context, a novel data analysis pattern is urgently expected to integrate effective data and to extract, store and learn information and manage knowledge for large-scale software projects. Due to its bright prospect of being used in various applications [1], software engineering-oriented data mining has become a major research topic in the fields of software engineering, data mining, artificial intelligence and pattern recognition.

## 2. Basic Concepts and Technological Challenges

Software engineering data refers to the broad variety of data accumulated during software development. It includes software codes, annotations, requirement documents, working instructions, test cases, test results, communications between developers, and user feed backs [4]. In most cases, software engineering data is the sole source of information for developers.

Due to the increase in the size of software, the volume and complexity of data that is generated during software development has been increasing quickly. Linux, a well-known open-source operating system, has approximately 6 million lines of codes. Hence, it is more difficult for developers to extract desired information from the engineering data by traditional means. This highlights the need to study the mining of software engineering data using data mining techniques.

Taking uniqueness of software engineering data into account, software engineering-oriented data mining confronts many unprecedented challenges. Details are as follows.

(1) Data complexity: software engineering data can be classified into two categories: structural data and non-structural data. The former refers to the data with fixed format, such as the defect report and the version information. The latter includes the codes, annotations and documents. The two categories of data should not adopt the same algorithm. Meanwhile, there is important correspondence between the two categories of data. For example, a defect report corresponds to the code segment that contains defects [5]. The structural data has some non-structural information, and the non-structural data also has some structural information. An example is the citation and link between software documents [6]. Furthermore, software engineering data has a broad variety of formats and structures. Novel algorithms are needed to mine data with such complexity.

(2) Non-traditional analysis: the final step and ultimate purpose of data mining is to deliver the obtained knowledge to users. In the traditional data mining fields, such as the e-commerce and banking industries, the knowledge is delivered to users in the form of texts and graphics. But the software developers need much more than statistical results. To facilitate software development, they also need specific cases, programming templates, defect locations and other information that describes the designed structure of software. Hence, it is necessary to propose new ways to deliver, display and assess knowledge obtained through data mining in the field of software engineering.

(3) Evaluation criterion of data mining results: mature systems are available to represent and evaluate data mining results in the traditional data mining applications. But this is not true for software engineering-oriented data mining. The information required by software developers is very specific and complicated, and it can be represented in various ways. It is thus difficult to make a comparison between these pieces of information and quantitatively analyze the data mining results.

To sum up, the ultimate purpose of data mining for software engineering is to acquire knowledge, and the difficulty lies in the pre-processing of data and the representation of knowledge

## 3. Status Quo of Data Mining Application in Software Engineering Projects

### *3.1 Reason behind widespread use of data mining techniques*

There are two reasons behind widespread use of data mining techniques. First, the rapidly increased memory capacity of the computer and the transmission bandwidth of the network enable the applications to store a large amount of data that can be analyzed. Second, due to continuous advances in modern information and software development techniques, in-depth studies have been carried out on many data mining algorithms. This is why these algorithms penetrate into specific fields unprecedentedly to extract implicit user needs. Furthermore, this type of technique enables the data to be stored easily and incurs a low application cost. The data mining technique is independent of other technological platforms, as it only needs to analyze existing data platforms in detail, resulting in reduced hardware cost. Hence, it has been attracting a lot of attention in the field of software project development.

### *3.2 The need for data mining techniques with greater performance*

In efforts to innovate the data mining techniques, the modern software engineering projects are more focused on the discovery of reusable analysis rules and the query of projects. The chains structure of the tree-like network and static/dynamic network rules are used to study the rules, mining the integrity of software engineering from the standpoint of rules. This further innovates the means of rule query, diversifies the form of queried rules and improves query efficiency [3]. In addition, the value of data mining cannot be embodied until it is used in practical applications. The broad variety of data mining algorithms that have been studied profoundly have their own application fields. For example, the Apriori algorithm is used for marketing, due to its ability to analyze existing data and make very accurate predictions.

The traditional defect positioning technique cannot locate defects of software projects completely. After combining with the data mining technique, the defect positioning system is more robust and more capable of representing rules. Software engineering is formulated as a model to better represent the behavior of the sequence and the tree and to automate evaluation. At the same time, it completes the projects on reverse engineering and automatic document generation, and improves the function of malicious software query. The more powerful data mining techniques are able to position the virus quickly, analyze and evaluate the virus data completely, as well as effectively improving the feasibility of data analysis and the security of software development. This converts studies on software engineering into a dynamic process.

## 4. Features of Data Mining

Although data mining in the field of software engineering has developed in different directions in recent years and the research in each direction has achieved remarkable results, it displays the following six distinct features.

(1) In most of the studies, software engineering-oriented data mining is formulated as a text mining problem.

(2) There are three types of data representation models: bag of words model, vector space model, and graph model.

(3) The major methods used in the studies consist of existing data mining and machine learning algorithms, such as the frequent sequential pattern algorithm, association rule, classification and clustering algorithms.

(4) The commonly used algorithms include the Apriori algorithm, naive Bayesian algorithm, decision tree classification algorithm, kNN algorithm and SVM.

(5) During pre-processing, data dimensionality is usually reduced using the methods commonly used in the field of text mining, such as SVD, LSI, PLSI and LDA.

(6) In addition to the commonly used static code and relevant documents, dynamic programs are adopted in some works to process the tracks and status information during data mining.

## 5. Future of Software Engineering-Oriented Data Mining

The ultimate purpose of data mining in the field of software engineering is to improve the efficiency and quality of software development. Software engineering involves purely intellectual work. To this end, the desired information should be fed to software developers efficiently, easily and accurately. Currently available tools for software engineering-oriented data mining cannot effectively meet the needs of software projects. Changes are always occurring to each element of software engineering-oriented data mining, such as the concepts and methods of software engineering, the structure and amount of software data, and the data mining algorithms. Due to these reasons, the study and application of software engineering-oriented data mining is also evolving. Breakthroughs are expected to be achieved in the following fields.

(1) Propose new software engineering-oriented data mining problems and solutions

① Reverse engineering and automatic document generation: existing research achievements are able to identify the correspondence between software requirement and design document [6] and the correspondence between general plan and requirement document [7]. In the future, efforts need to be taken to study reverse engineering and analyze software structure using data mining techniques.

② Identification of malicious software: we need to develop methods for software evaluation and malicious software identification. The method proposed in [8] can identify computer virus via data mining, but the malicious software is not limited to computer virus. It also includes the software that affects user experience and system security, such as the pop-up advertisement and the backdoor programs. It is more challenging to identify malicious software, as it needs to evaluate software behavior comprehensively.

③ Data mining-based software trustworthiness improvement and assessment: software trustworthiness is not equivalent to software correctness. Generally speaking, software trustworthiness includes software correctness, reliability and security. Defect in software trustworthiness usually affects software correctness. By improving software reliability and security via data mining, we can enable the software to operate more stably in the extreme environment, and identify and resist various attacks from malicious users more effectively. But few works have been done on this issue. Although it is significant to prove software trustworthiness in some aspect by formal means [117], the application scope is limited. If we obtain statistical conclusions by quantitatively assessing and predicting software trustworthiness through data mining, we can offer guidance on software development and maintenance.

(2) Algorithm for efficient pre-processing and mining of the large amount of software engineering data. The size of software engineering data has been on the rise in the context of increased demand for software, extension of software development tools, increased size of software development teams, and the lengthened cycle of software development. However, only a small amount of data is mined currently in the field of software engineering, as the mined data is merely a part of the large-scale software database. For example, in the studies about repetitious defect detection report [10], only a small proportion (less than 1%) is extracted from the library of defect reports of the Eclipse and Mozilla open-source software. This contributes to inefficiency and poor suitability of existing algorithms. To address this problem, we need to develop and adopt more efficient data pre-processing methods, especially the feature extraction and dimensionality reduction algorithms. On the other hand, we need to develop more efficient data mining algorithms, including the algorithm for online mining of dynamic data streams.

In addition, the software development organizations are more international and open. A software development team can be distributed across the world, adopting different working languages and devices as well as distinctive development techniques and standards. Hence, the large amount of software engineering data is heterogeneous and distributed. In this context, the studies on software engineering-oriented data mining inevitably involves the management and fusion of heterogeneous data (e.g., data fusion and data space) and parallel processing of distributed data (e.g., MapReduce). XX provides a very feasible approach to this problem.

(5) Develop novel and more efficient software engineering-oriented data mining algorithms by making full

use of latest advances in artificial intelligence, machine learning and data mining. It has been stated above that the software engineering data is currently mined using the mature but old machine learning and data mining methods. But remarkable progress has been achieved in machine learning during the past 10 years, yielding many new algorithms, such as ensemble learning, manifold learning, active learning, transfer learning, and the probabilistic graphical model. Introducing these new methods to software engineering-oriented data mining will yield new and more efficient data mining algorithms.

To sum up, excellent progress will be made on the use of data mining techniques for software engineering projects in the near future. And the software engineering projects will be more intelligent and automatic. From the research progress, it can be seen that there is enormous room for development of software engineering and data mining. The technical tools will also be enhanced to create more value.

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